

Lecture 13

Blade dynamic response

Part 1

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This lecture

- Modal synthesis
- Steady-state flapping response
- Uncoupled flapping oscillators
- F1 mode oscillator example

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Periodic and harmonic blade excitation

Previously completed eigenvalue (or modal) analysis is closely linked to the free response of rotating blades in vacuum. The next study considers the steady-state response of blades under **periodic excitation** due to the distributed lift $L(x,t)$.

The governing equation of the rotating flapping blade with forcing, $L(x,t) \neq 0$:

$$(EI w'')'' - \Omega^2 (\tilde{T} w')' + m\ddot{w} = L$$

Rotor excitation during steady flight can be expressed as follows:

$$L(x,t) = \sum_{(k)} L_k(x) e^{ik\Omega t} = L_0(x) + \sum_{(k)} L_k(x) \sin(k\Omega t + \varphi_k)$$

$$L(x,t) = L_0(x) + \sum_{(k)} L_{S,k}(x) \sin(k\Omega t) + L_{C,k}(x) \cos(k\Omega t)$$



Consider the case of **harmonic forcing** at frequency $k\Omega$:

$$L(x,t) = L_k(x) \sin(k\Omega t)$$

Later we'll consider $k=1,2, \dots$

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Modal synthesis

Modal synthesis is used to solve the problem with the forcing. The main idea is to express the blade's flapping response $w(x,t)$ as a sum of the mode shapes multiplied by the time-dependent weighting coefficients called the modal participation factors.

Flapping blade equation with forcing, $L(x,t) \neq 0$

$$(EI w'')'' - \Omega^2 (\tilde{T} w')' + m\ddot{w} = L$$

Variable separation: $w(x,t) = Q(t)W(x)$

$$w(x,t) = q_1(t) \times w_1(x) + q_2(t) \times w_2(x) + q_3(t) \times w_3(x) + \dots$$

General blade deflection expressed as the sum of the **modal contributions**:

$$w(x,t) = q_1 w_1 + q_2 w_2 + q_3 w_3 + \dots = \sum_{i=1}^{\infty} q_i(t) w_i(x)$$

All possible blade deflections are written as the sum of the modal contributions, where $w_i(x)$ are the mode shape functions and $q_i(t)$ are the **mode participation factors**.

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Steady-state response using modal synthesis

Determine the state-state response under harmonic excitation:

$$(EI w'')'' - \Omega^2 (\tilde{T} w')' + m \ddot{w} = L_k(x) \sin(k\Omega t)$$

$$w(x, t) = \sum_{i=1}^{\infty} q_i(t) w_i(x) = \sum q_i w_i$$

Substitute the proposed $w(x, t)$ and differentiate as required:

$$(EI (\sum q_i w_i)''')'' - \Omega^2 (\tilde{T} (\sum q_i w_i)')' + m d^2(\sum q_i w_i)/dt^2 = L_k \sin(k\Omega t)$$

$$\sum q_i (EI (w_i)''')'' - \Omega^2 \sum q_i (\tilde{T} w_i')' + \sum \ddot{q}_i (m w_i) = L_k \sin(k\Omega t)$$

Multiply the above equation by the j -th mode $w_j(x)$ and integrate the equation along the blade span:

$$\sum q_i \int_{(b)} (EI (w_i)''')'' w_j dx - \Omega^2 \sum q_i \int_{(b)} (\tilde{T} w_i')' w_j dx + \sum \ddot{q}_i \int_{(b)} m w_i w_j dx = \int_{(b)} w_j L_k dx \sin(k\Omega t)$$

$$= \int_{(b)} w_j L_k dx \sin(k\Omega t)$$

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Steady-state response using modal synthesis, cont.

Apply the same steps as in the derivation of the Southwell's formulas:

$$\sum q_i \left(\int_{(b)} (EI (w_i)''')'' w_j dx - \Omega^2 \int_{(b)} (\tilde{T} w_i')' w_j dx \right) + \sum \ddot{q}_i \int_{(b)} m w_i w_j dx = \int_{(b)} w_j L_k dx \sin(k\Omega t)$$

Use integration *by parts* and apply the standard blade BCs (hinge, fixed, free):

$$\sum q_i \left(\int_{(b)} EI w_i'' w_j'' dx + \Omega^2 \int_{(b)} \tilde{T} w_i' w_j' dx \right) + \sum \ddot{q}_i \int_{(b)} m w_i w_j dx = \int_{(b)} w_j L_k dx \sin(k\Omega t)$$

It can be shown that when $i \neq j$, the corresponding integrals are zero. Then, only the terms where $i = j$ are retained and the resulting equation is:

$$q_j \left(\underbrace{\int_{(b)} EI w_j''^2 dx}_{\text{Modal stiffness } K_j} + \underbrace{\Omega^2 \int_{(b)} \tilde{T} w_j'^2 dx}_{\text{Modal mass } M_j} \right) + \underbrace{\ddot{q}_j \int_{(b)} m w_j^2 dx}_{\text{Modal mass } M_j} = \underbrace{\left(\int_{(b)} w_j L_k dx \right)}_{\text{Modal load } L_{j,k}} \sin(k\Omega t)$$

Finally, rearrange the equation to the standard ODE form:

$$\ddot{q}_j + \left(\underbrace{\int_{(b)} EI w_j''^2 dx / M_j + \Omega^2 \int_{(b)} \tilde{T} w_j'^2 dx / M_j}_{\omega_{F,j}^2 = \omega_{0,F,j}^2 + \Omega^2 K_{F,j}} \right) q_j = \left(\int_{(b)} w_j L_k dx / M_j \right) \sin(k\Omega t)$$

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Steady-state blade response – summary

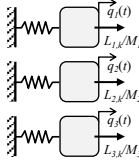
Solve for the state-state response of the system subjected to harmonic excitation at $k\Omega$:

$$(EI w'')'' - \Omega^2 (\tilde{T} w')' + m\ddot{w} = L_k(x) \sin(k\Omega t) \quad w(x,t) = \sum_{j=1}^{\infty} q_j(t) w_j(x)$$

The evolution of each modal participation factor function is determined by the following second-order ODE subjected to the harmonic excitation at $k\Omega$:

$$\ddot{q}_j + (\omega_{0,F,j}^2 + \Omega^2 K_{F,j}) q_j = (L_{j,k} / M_j) \sin(k\Omega t), \quad j = 1, 2, \dots$$

The original IV-BVP PDE is transformed to a system of ODEs:

$$\begin{aligned} \ddot{q}_1 + \omega_{F,1}^2 q_1 &= (L_{1,k} / M_1) \sin(k\Omega t) \\ \ddot{q}_2 + \omega_{F,2}^2 q_2 &= (L_{2,k} / M_2) \sin(k\Omega t) \\ \ddot{q}_3 + \omega_{F,3}^2 q_3 &= (L_{3,k} / M_3) \sin(k\Omega t) \\ &\dots \end{aligned}$$


Each ODE represents an uncoupled 2nd order oscillator, here, with harmonic excitation at $k\Omega$. The individual solutions determine the contribution of the j -th mode to $w(x,t)$. Initial conditions are not required when solving for the steady-state $w(x,t)$.

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Example: blade response using F1 mode

Assume a blade with $EI=5469 \text{ Nm}^2$, $m=4.3 \text{ kg/m}$ and $R=5 \text{ m}$. Use the previously determined assumed **quartic polynomial representation of mode F1** to describe the overall flapping response of the rotating blade. Derive the coefficients and write the ODE for the corresponding mode participation factor.

$$\ddot{q}_1 + q_1 \omega_{F,1}^2 = \frac{L_{1,k}}{M_1} \sin(k\Omega t) \quad \rightarrow \quad w(x,t) = \sum_{i=1}^1 q_i(t) w_i(x)$$

The assumed F1 function is used to complete the blade integrals and determine the ODE coefficients:

$$f(x) = \frac{1}{3} \left(\frac{x}{R} \right)^4 - \frac{4}{3} \left(\frac{x}{R} \right)^3 + 2 \left(\frac{x}{R} \right)^2 \quad \left\{ \begin{aligned} M_1 &\approx m \int_0^R f^2 dx = 5.521 \text{ kg} \\ \omega_{F,1} &\approx 5.036 \text{ rad/s} + \sqrt{1.173} \Omega \\ L_{1,k} &\approx \int_0^5 f(x) L_k(x) dx \end{aligned} \right.$$

The lift distribution is not given here. Such term would require introduction of aerodynamic model to determine $L(x,t)$ for each time-integration step.

The modal participation factor q_1 is determined then by the following:

$$\ddot{q}_1 + (5.036^2 + 1.173\Omega^2) q_1 = \frac{1}{5.521} \left(\int_0^5 f(x) L_k(x) dx \right) \sin(k\Omega t)$$

After that, the blade response (deformations, moments, etc.) is:

$$w(x,t) \approx q_1(t) f(x), \quad M(x,t) \approx EI(x) (q_1(t) f''(x))$$

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Summary

- Modal synthesis is a form of variable separation strategy used to transform the governing (field) PDE to a set of uncoupled second-order ODEs in mode participation factors.
- The eigenvalue BVP is used to obtain the vibration modes and the ODE system is used to determine the participation of the modes on the response under given excitation.
- The steady-state responses are typically sought and studied when considering the rotor-induced periodic blade loading.
- The dynamic blade response problem is expressed as a system of modal ODEs with periodic excitation represented as a sum of a few rotor-frequency harmonic components.

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